

§ 4—1

1. 填空题

(1) 设 $f(x)$ 连续,

$$\text{则 } d\int f(x)dx = \underline{f(x)dx};$$

$$\int df(x) = \underline{f(x)+C};$$

$$\frac{d}{dx}\int f(x)dx = \underline{f(x)};$$

$$\int f'(x)dx = \underline{f(x)+C}.$$

(2) 设 $F_1(x)$, $F_2(x)$ 是 $f(x)$

的两个不同的原函数,

且 $f(x) \neq 0$,

$$\text{则 } F_1(x) - F_2(x) = \underline{C}.$$

$$(3) \int \frac{3+x^2}{1+x^2} dx =$$

$$\underline{2\arctan x + x + C}.$$

$$\text{解: } \int \frac{3+x^2}{1+x^2} dx = \int \frac{2+(1+x^2)}{1+x^2} dx$$

$$= 2\arctan x + x + C.$$

$$(4) \int \left(\frac{4}{\sqrt{1-x^2}} dx + \frac{3}{\sqrt{x}} \right) dx =$$

$$\underline{4\arcsin x + 6\sqrt{x} + C}.$$

$$(5) \int (2^x + 3^x)^2 dx =$$

$$\underline{\frac{4^x}{\ln 4} + 2 \cdot \frac{6^x}{\ln 6} + \frac{9^x}{\ln 9} + C}.$$

$$\text{解: } \int (2^x + 3^x)^2 dx$$

$$= \left[\int (2^x)^2 + 2 \cdot 2^x \cdot 3^x + (3^x)^2 \right] dx$$

$$= \int (4^x + 2 \cdot 6^x + 9^x) dx$$

$$= \frac{4^x}{\ln 4} + 2 \cdot \frac{6^x}{\ln 6} + \frac{9^x}{\ln 9} + C.$$

$$(6) \int \sec x (\sec x - \tan x) dx =$$

$$\underline{\tan x - \sec x + C}.$$

$$\text{解: } \int \sec x (\sec x - \tan x) dx$$

$$= \int (\sec^2 x - \sec x \tan x) dx$$

$$= \tan x - \sec x + C.$$

$$(7) \int \frac{e^{2x} - 1}{e^x - 1} dx = \underline{e^x + x + C}.$$

$$\text{解: } \int \frac{e^{2x} - 1}{e^x - 1} dx = \int \frac{(e^x)^2 - 1}{e^x - 1} dx$$

$$= \int (e^x + 1) dx = e^x + x + C.$$

2. 已知 $F(x)$ 是 $\frac{\ln x}{x}$ 的一个

原函数, 求 $dF(\sin x)$.

解: 根据已知, 有

$$F'(x) = \frac{\ln x}{x}. \text{ 则}$$

$$dF(\sin x) = F'(\sin x) d(\sin x)$$

$$= \frac{\ln \sin x}{\sin x} \cdot \cos x dx$$

$$= \cot x \cdot \ln \sin x dx.$$

§ 4—2

1. 填空题

$$(1) \int \frac{dx}{\sqrt{3-2x}} = \underline{-\sqrt{3-2x} + C}.$$

$$\text{解: } \int \frac{dx}{\sqrt{3-2x}} = -\frac{1}{2} \int \frac{d(3-2x)}{\sqrt{3-2x}}$$

$$= -\frac{1}{2} \cdot 2\sqrt{3-2x} + C$$

$$= -\sqrt{3-2x} + C.$$

$$(2) \int e^x \sin e^x dx = \underline{-\cos e^x + C}.$$

$$\begin{aligned}\text{解: } \int e^x \sin e^x dx &= \int \sin e^x de^x \\ &= -\cos e^x + C.\end{aligned}$$

$$(3) \int \frac{1}{x^2} \cos \frac{1}{x} dx = \underline{-\sin \frac{1}{x} + C}.$$

$$\begin{aligned}\text{解: } \int \frac{1}{x^2} \cos \frac{1}{x} dx \\ &= -\int \cos \frac{1}{x} d\left(\frac{1}{x}\right) \\ &= -\sin \frac{1}{x} + C.\end{aligned}$$

$$(4) \int \frac{1}{x \ln x} dx = \underline{\ln |\ln x| + C}.$$

$$\text{解: 原式} = \int \frac{d \ln x}{\ln x} = \ln |\ln x| + C.$$

$$(5) \int x^2 \sqrt{1+x^3} dx = \underline{\frac{2}{9}(1+x^3)^{\frac{3}{2}} + C}.$$

$$\begin{aligned}\text{解: } \int x^2 \sqrt{1+x^3} dx &= \frac{1}{3} \int \sqrt{1+x^3} dx^3 \\ &= \frac{1}{3} \cdot \frac{2}{3} (1+x^3)^{\frac{3}{2}} + C \\ &= \frac{2}{9} (1+x^3)^{\frac{3}{2}} + C.\end{aligned}$$

$$(6) \int \left(1 - \frac{1}{x^2}\right) e^{x+\frac{1}{x}} dx = \underline{e^{x+\frac{1}{x}} + C}.$$

$$\begin{aligned}\text{解: } \int \left(1 - \frac{1}{x^2}\right) e^{x+\frac{1}{x}} dx \\ &= \int e^{x+\frac{1}{x}} d\left(x + \frac{1}{x}\right) = e^{x+\frac{1}{x}} + C.\end{aligned}$$

$$(7) \int \frac{dx}{1+9x^2} = \underline{\frac{1}{3} \arctan(3x) + C}.$$

$$\text{解: } \int \frac{dx}{1+9x^2} = \frac{1}{3} \int \frac{d(3x)}{1+(3x)^2}.$$

$$= \frac{1}{3} \arctan(3x) + C$$

$$(8) \int \frac{dx}{\sqrt{4-9x^2}} = \frac{1}{3} \arcsin\left(\frac{3x}{2}\right) + C.$$

$$\text{解: } \int \frac{dx}{\sqrt{4-9x^2}} = \frac{1}{3} \int \frac{d(3x)}{\sqrt{2^2-(3x)^2}}$$

$$= \frac{1}{3} \arcsin\left(\frac{3x}{2}\right) + C.$$

2. 计算下列各题:

$$(1) \int \frac{\tan x}{\sqrt{\cos x}} dx.$$

$$\text{解: } \int \frac{\tan x}{\sqrt{\cos x}} dx = \int \frac{\sin x}{\cos x \sqrt{\cos x}} dx$$

$$= -\int (\cos x)^{-\frac{3}{2}} d(\cos x)$$

$$= 2(\cos x)^{-\frac{3}{2}+1} + C$$

$$= 2(\cos x)^{-\frac{1}{2}} + C = \frac{2}{\sqrt{\cos x}} + C.$$

$$(2) \int \tan^3 x \sec x dx.$$

$$\text{解: } \int \tan^3 x \sec x dx$$

$$= \int \tan^2 x \cdot \tan x \sec x dx$$

$$= \int (\sec^2 x - 1) d(\sec x)$$

$$= \frac{\sec^3 x}{3} - \sec x + C.$$

$$(3) \int \frac{dx}{e^x + e^{-x}}.$$

$$\text{解: } \int \frac{dx}{e^x + e^{-x}} = \int \frac{dx}{e^{-x}(e^{2x} + 1)}$$

$$= \int \frac{d(e^x)}{1+(e^x)^2} = \arctan(e^x) + C.$$

$$\text{解法二: } \int \frac{dx}{e^x + e^{-x}} = \int \frac{dx}{e^x(1+e^{-2x})}$$

$$= -\int \frac{d(e^{-x})}{1+(e^{-x})^2} = -\arctan(e^{-x}) + C.$$

$$(4) \int \frac{1+\ln x}{(x \ln x)^2} dx.$$

$$\text{解: } \int \frac{1+\ln x}{(x \ln x)^2} dx$$

$$= \int \frac{1}{(x \ln x)^2} d(x \ln x)$$

$$= -\frac{1}{x \ln x} + C.$$

$$(5) \int \frac{1}{\sqrt{x}} \sin \sqrt{x} dx.$$

$$\text{解: } \int \frac{1}{\sqrt{x}} \sin \sqrt{x} dx$$

$$= 2 \int \sin \sqrt{x} d\sqrt{x} = -2 \cos \sqrt{x} + C.$$

$$(6) \int \frac{dx}{\cos^2 x \sqrt{1+\tan x}}.$$

$$\text{解: 原式} = \int \frac{d \tan x}{\sqrt{1+\tan x}}$$

$$= \int \frac{d(1+\tan x)}{\sqrt{1+\tan x}} = 2\sqrt{1+\tan x} + C.$$

$$(7) \int \frac{\sin x \cos x}{1+\cos^2 x} dx.$$

$$\text{解: } \int \frac{\sin x \cos x}{1+\cos^2 x} dx$$

$$= -\int \frac{\cos x}{1+\cos^2 x} d(\cos x)$$

$$\begin{aligned}
&= -\frac{1}{2} \int \frac{d(\cos^2 x)}{1 + \cos^2 x} \\
&= -\frac{1}{2} \int \frac{d(1 + \cos^2 x)}{1 + \cos^2 x} \\
&= -\frac{1}{2} \ln(1 + \cos^2 x) + C.
\end{aligned}$$

解法二: $\int \frac{\sin x \cos x}{1 + \cos^2 x} dx$

$$\begin{aligned}
&= \frac{1}{2} \int \frac{\sin 2x}{1 + \frac{\cos 2x}{2}} dx \\
&= \int \frac{\sin 2x}{3 + \cos 2x} dx \\
&= -\frac{1}{2} \int \frac{d(3 + \cos 2x)}{3 + \cos 2x} \\
&= -\frac{1}{2} \ln(3 + \cos 2x) + C.
\end{aligned}$$

3. 计算下列不定积分:

(1) $\int \frac{dx}{(x^2 + 1)^{\frac{3}{2}}}.$

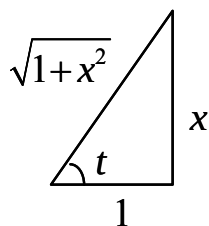
解: (三角代换)

令 $x = \tan t \left(0 < t < \frac{\pi}{2} \right),$

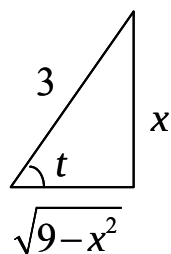
则 $dx = \sec^2 t dt.$

于是 $\int \frac{dx}{(x^2 + 1)^{\frac{3}{2}}} = \int \frac{\sec^2 t}{\sec^3 t} dt$

$$\begin{aligned}
&= \int \frac{1}{\sec t} dt = \int \cos t dt \\
&= \sin t + C = \frac{x}{\sqrt{1 + x^2}} + C.
\end{aligned}$$



$$(2) \int \frac{\sqrt{9-x^2}}{x} dx.$$



解：令 $x = 3 \sin t$,

$-\frac{\pi}{2} < t < \frac{\pi}{2}$, 则

$$\int \frac{\sqrt{9-x^2}}{x} dx$$

$$= \int \frac{\sqrt{9-(3\sin t)^2}}{3\sin t} d(3\sin t)$$

$$= \int \frac{3\cos t \cdot 3\cos t}{3\sin t} dt$$

$$= 3 \int \frac{\cos^2 t}{\sin t} dt = 3 \int \frac{1-\sin^2 t}{\sin t} dt$$

$$= 3 \int \left(\frac{1}{\sin t} - \sin t \right) dt$$

$$= 3 \int (\csc t - \sin t) dt$$

$$= 3 \ln |\csc t - \cot t| + 3 \cos t + 3C$$

$$= 3 \ln \left| \frac{1}{\sin t} - \cot t \right| + 3 \cos t + C_1$$

$$(C_1 = 3C)$$

$$= 3 \ln \left| \frac{3}{x} - \frac{\sqrt{9-x^2}}{x} \right| + 3 \cdot \frac{\sqrt{9-x^2}}{3} + C_1$$

$$= 3 \ln \left| \frac{3 - \sqrt{9 - x^2}}{x} \right| + \sqrt{9 - x^2} + C_1.$$

4. 已知 $f(x)$ 的一个原函数为

$$\frac{\sin x}{1 + x \sin x}, \text{ 求 } \int f(x) f'(x) dx.$$

解: 根据已知, 有

$$\begin{aligned} f(x) &= \left(\frac{\sin x}{1 + x \sin x} \right)' = \\ &= \frac{(\sin x)' \cdot (1 + x \sin x) - \sin x \cdot (1 + x \sin x)'}{(1 + x \sin x)^2} \\ &= \frac{\cos x \cdot (1 + x \sin x) - \sin x \cdot (\sin x + x \cos x)}{(1 + x \sin x)^2} \\ &= \frac{\cos x - \sin^2 x}{(1 + x \sin x)^2}. \end{aligned}$$

所以

$$\begin{aligned} \int f(x) f'(x) dx &= \int f(x) df(x) \\ &= \frac{1}{2} [f(x)]^2 + C \\ &= \frac{1}{2} \left[\frac{\cos x - \sin^2 x}{(1 + x \sin x)^2} \right]^2 + C. \end{aligned}$$

§ 4—3

1. 计算下列不定积分:

$$(1) \int x \sin^2 x dx.$$

$$\begin{aligned} \text{解: 原式} &= \int x \cdot \frac{1 - \cos 2x}{2} dx \\ &= \frac{1}{2} \int x dx - \frac{1}{2} \int x \cos 2x dx \\ &= \frac{1}{4} x^2 - \frac{1}{4} \int x d(\sin 2x) \\ &= \frac{1}{4} x^2 - \frac{1}{4} x \sin 2x + \frac{1}{4} \int \sin 2x dx \\ &= \frac{1}{4} x^2 - \frac{1}{4} x \sin 2x - \frac{1}{8} \cos 2x + C. \end{aligned}$$

或 原式

$$\begin{aligned} &= \frac{x^2}{2} \sin^2 x + \frac{x^2}{4} \cos 2x \\ &\quad - \frac{x}{4} \sin 2x - \frac{1}{8} \cos 2x + C. \end{aligned}$$

$$(2) \int \arctan \sqrt{x} dx .$$

解： 设 $\sqrt{x} = t$,

则 $x = t^2$, $dx = 2tdt$.

于是 $\int \arctan \sqrt{x} dx = 2 \int t \arctan t dt$

$$= \int \arctan t d(t^2)$$

$$= t^2 \arctan t - \int \frac{t^2}{1+t^2} dt$$

$$= t^2 \arctan t - \int \frac{t^2 + 1 - 1}{1+t^2} dt$$

$$= t^2 \arctan t - \int \left(1 - \frac{1}{1+t^2} \right) dt$$

$$= t^2 \arctan t - t - \arctan t + C$$

$$= x \arctan \sqrt{x} - \sqrt{x} - \arctan \sqrt{x} + C .$$

解法二： $\int \arctan \sqrt{x} dx$

$$= x \arctan \sqrt{x} - \int x d(\arctan \sqrt{x})$$

$$= x \arctan \sqrt{x} - \int \frac{x}{1+(\sqrt{x})^2} d(\sqrt{x})$$

$$= x \arctan \sqrt{x} - \int \frac{x+1-1}{1+x} d(\sqrt{x})$$

$$= x \arctan \sqrt{x} - \int \left(1 - \frac{1}{1+x} \right) d(\sqrt{x})$$

$$= x \arctan \sqrt{x}$$

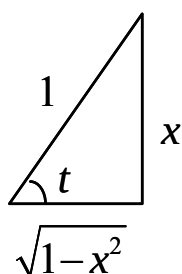
$$- \int d(\sqrt{x}) + \int \frac{1}{1+\sqrt{(x)^2}} d(\sqrt{x})$$

$$= x \arctan \sqrt{x} - \sqrt{x} + \arctan \sqrt{x} + C .$$

$$(3) \int \frac{\arcsin x}{x^2} dx .$$

$$\begin{aligned}
 \text{解: 原式} &= -\int \arcsin x d\left(\frac{1}{x}\right) \\
 &= -\frac{1}{x} \arcsin x + \int \frac{1}{x} d(\arcsin x) \\
 &= -\frac{1}{x} \arcsin x + \int \frac{1}{x} \cdot \frac{1}{\sqrt{1-x^2}} dx \\
 &= -\frac{1}{x} \arcsin x + \int \frac{1}{x\sqrt{1-x^2}} dx.
 \end{aligned}$$

$$\text{令 } x = \sin t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2},$$



则

$$\begin{aligned}
 \int \frac{1}{x\sqrt{1-x^2}} dx &= \int \frac{1}{\sin t \cos t} d(\sin t) \\
 &= \int \frac{\cos t}{\sin t \cos t} dt = \int \frac{1}{\sin t} dt \\
 &= \int \csc t dt = \ln |\csc t - \cot t| + C \\
 &= \ln \left| \frac{1}{\sin t} - \cot t \right| + C \\
 &= \ln \left| \frac{1}{x} - \frac{\sqrt{1-x^2}}{x} \right| + C,
 \end{aligned}$$

所以

$$\begin{aligned}
 &\int \frac{\arcsin x}{x^2} dx \\
 &= -\frac{1}{x} \arcsin x + \ln \left| \frac{1}{x} - \frac{\sqrt{1-x^2}}{x} \right| + C.
 \end{aligned}$$

$$(4) \int e^{-x} \sin 2x dx$$

$$\text{解: } \int e^x \sin 2x dx = -\int \sin 2x d(e^{-x})$$

$$= -e^{-x} \sin 2x + \int e^{-x} d(\sin 2x)$$

$$= -e^{-x} \sin 2x + 2 \int e^{-x} \cos 2x dx$$

$$= -e^{-x} \sin 2x - 2 \int \cos 2x d(e^{-x})$$

$$= -e^{-x} \sin 2x - 2(e^{-x} \cos 2x - \int e^{-x} d \cos 2x)$$

$$= -e^{-x} \sin 2x - 2e^{-x} \cos 2x - 4 \int e^{-x} \sin 2x dx .$$

移项合并, 得

$$\int e^x \sin 2x dx .$$

$$= -\frac{1}{5} e^{-x} (\sin 2x + 2 \cos 2x) + C .$$

$$(5) \int \frac{\arctan x}{x^2(1+x^2)} dx .$$

解: 原式

$$= \int \left(\frac{1}{x^2} - \frac{1}{1+x^2} \right) \arctan x dx$$

$$= \int \frac{\arctan x}{x^2} dx - \int \frac{\arctan x}{1+x^2} dx$$

$$= -\int \arctan x d\left(\frac{1}{x}\right)$$

$$-\int \arctan x d(\arctan x)$$

$$= -\frac{1}{x} \arctan x + \int \frac{1}{x} d(\arctan x)$$

$$-\frac{1}{2} (\arctan x)^2$$

$$= -\frac{1}{x} \arctan x + \int \frac{1}{x(1+x^2)} dx$$

$$-\frac{1}{2} (\arctan x)^2$$

$$= -\frac{1}{x} \arctan x + \int \left(\frac{1}{x} - \frac{x}{1+x^2} \right) dx$$

$$\begin{aligned}
& -\frac{1}{2}(\arctan x)^2 \\
& = -\frac{1}{x}\arctan x + \int \frac{1}{x} dx \\
& -\int \frac{x}{1+x^2} dx - \frac{1}{2}(\arctan x)^2 \\
& = -\frac{1}{x}\arctan x + \int \frac{1}{x} dx \\
& -\frac{1}{2}\int \frac{1}{1+x^2} d(1+x^2) - \frac{1}{2}(\arctan x)^2 \\
& = -\frac{1}{x}\arctan x + \ln|x| - \frac{1}{2}\ln(1+x^2) \\
& -\frac{1}{2}(\arctan x)^2 + C \\
& = -\frac{1}{x}\arctan x - \frac{1}{2}(\arctan x)^2 . \\
& -\frac{1}{2}\ln \frac{1+x^2}{x^2} + C .
\end{aligned}$$

2. 已知 $f(x)$ 的一个原函数是

$$\ln(x + \sqrt{1+x^2}), \text{ 求 } \int x f'(x) dx .$$

解: 根据已知, 有

$$\begin{aligned}
f(x) &= \left[\ln(x + \sqrt{1+x^2}) \right]' \\
&= \frac{1 + \frac{2x}{2\sqrt{1+x^2}}}{x + \sqrt{1+x^2}} = \frac{1 + \frac{x}{\sqrt{1+x^2}}}{x + \sqrt{1+x^2}} \\
&= \frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2}(x + \sqrt{1+x^2})} \\
&= \frac{1}{\sqrt{1+x^2}} .
\end{aligned}$$

由分部积分公式, 得

$$\begin{aligned}
\int x f'(x) dx &= \int x df(x) \\
&= x f(x) - \int f(x) dx \\
&= x \cdot \frac{1}{\sqrt{1+x^2}} - \int \frac{1}{\sqrt{1+x^2}} dx \\
&= \frac{x}{\sqrt{1+x^2}} - \ln(x + \sqrt{1+x^2}) + C .
\end{aligned}$$

3. 设 $f'(e^x) = 1+x$, 求 $f(x)$.

解：令 $e^x = t$ ，则 $x = \ln t$ ，

则 $f'(t) = 1 + \ln t$ 。

$$f(t) = \int f'(t)dt = \int (1 + \ln t)dt$$

$$= t(1 + \ln t) - \int t d(1 + \ln t)$$

$$= t(1 + \ln t) - \int t \cdot \frac{1}{t} dt$$

$$= t(1 + \ln t) - t + C$$

$$= t \ln t + C,$$

所以 $f(x) = x \ln x + C$ 。

§ 4—4

1. 求下列不定积分：

$$(1) \int \frac{x-5}{x^2+2x-3} dx.$$

$$\text{解：原式} = \frac{1}{2} \int \frac{2x+2-2-10}{x^2+2x-3} dx$$

$$= \frac{1}{2} \int \frac{d(x^2+2x-3)}{x^2+2x-3}$$

$$- \int \frac{6}{x^2+2x-3} dx \quad (\Delta > 0)$$

$$= \frac{1}{2} \ln |x^2+2x-3|$$

$$- 6 \int \frac{1}{(x+3)(x-1)} dx$$

$$= \frac{1}{2} \ln |x^2+2x-3|$$

$$+ \frac{6}{4} \left(\int \frac{1}{x+3} - \frac{1}{x-1} \right) dx$$

$$= \frac{1}{2} \ln |(x+3)(x-1)|$$

$$+ \frac{3}{2} \left[\int \frac{d(x+3)}{x+3} - \int \frac{d(x-1)}{x-1} \right]$$

$$= \frac{1}{2} (\ln |x+3| + \ln |x-1|)$$

$$+\frac{3}{2}(\ln|x+3|-\ln|x-1|)+C$$
$$=2\ln|x+3|-\ln|x-1|+C.$$

$$(2) \int \frac{x-5}{x^2+2x+3} dx.$$

$$\text{解: 原式} = \frac{1}{2} \int \frac{2x+2-2-10}{x^2+2x+3} dx$$

$$= \frac{1}{2} \int \frac{d(x^2+2x+3)}{x^2+2x+3}$$

$$-\int \frac{6}{x^2+2x+3} dx \quad (\Delta < 0)$$

$$= \frac{1}{2} \ln|x^2+2x+3|$$

$$-6 \int \frac{1}{(x+1)^2+2} dx$$

$$= \frac{1}{2} \ln|x^2+2x+3|$$

$$-6 \int \frac{d(x+1)}{(x+1)^2+(\sqrt{2})^2}$$

$$= \frac{1}{2} \ln|x^2+2x+3|$$

$$-\frac{6}{\sqrt{2}} \arctan \frac{x+1}{\sqrt{2}} + C$$

$$= \frac{1}{2} \ln|x^2+2x+3|$$

$$-3\sqrt{2} \arctan \frac{x+1}{\sqrt{2}} + C.$$

$$(3) \int \frac{dx}{(x^2+1)(x^2+x+1)}.$$

$$\text{解: 原式} = \int \frac{1}{x} \left(\frac{1}{x^2+1} - \frac{1}{x^2+x+1} \right) dx$$

$$= \int \frac{1}{x(x^2+1)} dx - \int \frac{1}{x \cdot (x^2+x+1)} dx$$

$$= \int \left(\frac{1}{x} - \frac{x}{x^2+1} \right) dx - \int \left(\frac{1}{x} - \frac{x+1}{x^2+x+1} \right) dx$$

$$\begin{aligned}
&= -\frac{1}{2} \int \frac{d(x^2+1)}{x^2+1} + \frac{1}{2} \int \frac{d(x^2+x+1)}{x^2+x+1} \\
&\quad + \frac{1}{2} \int \frac{1}{x^2+x+1} dx \quad (\Delta < 0) \\
&= -\frac{1}{2} \ln(x^2+1) + \frac{1}{2} \ln|x^2+x+1| \\
&\quad + \frac{1}{2} \int \frac{1}{\left(x+\frac{1}{2}\right)^2 + \frac{3}{4}} dx \\
&= -\frac{1}{2} \ln(x^2+1) + \frac{1}{2} \ln|x^2+x+1| \\
&\quad + \frac{1}{2} \int \frac{d\left(x+\frac{1}{2}\right)}{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\
&= -\frac{1}{2} \ln(x^2+1) + \frac{1}{2} \ln|x^2+x+1| \\
&\quad + \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \arctan \frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}} + C \\
&= -\frac{1}{2} \ln(x^2+1) + \frac{1}{2} \ln|x^2+x+1| \\
&\quad + \frac{\sqrt{3}}{3} \arctan \frac{2x+1}{\sqrt{3}} + C.
\end{aligned}$$

$$(4) \int \frac{1}{x(x^{10}-2)} dx.$$

$$\begin{aligned}
\text{解: } \int \frac{1}{x(x^{10}-2)} dx &= \int \frac{x^9}{x^{10}(x^{10}-2)} dx \\
&= \frac{1}{10} \int \frac{1}{x^{10}(x^{10}-2)} d(x^{10}) \\
&= -\frac{1}{20} \int \left(\frac{1}{x^{10}} - \frac{1}{x^{10}-2} \right) d(x^{10}) \\
&= -\frac{1}{20} \int \frac{1}{x^{10}} d(x^{10})
\end{aligned}$$

$$\begin{aligned}& + \frac{1}{20} \int \frac{1}{x^{10}-2} d(x^{10}-2) \\& = -\frac{1}{20} \ln(x^{10}) + \frac{1}{20} \ln|x^{10}-2| + C. \\& = -\frac{1}{20} \ln \left| \frac{x^{10}-2}{x^{10}} \right| + C.\end{aligned}$$

$$(5) \int x\sqrt{1-2x}dx.$$

解: 设 $\sqrt{1-2x}=t$, 则 $1-2x=t^2$,

$$x = \frac{1-t^2}{2}, \quad dx = -tdt. \text{ 于是}$$

$$\begin{aligned}\int x\sqrt{1-2x}dx &= \int \frac{1-t^2}{2} \cdot t \cdot (-t)dt \\&= \frac{1}{2} \int (1-t^2) \cdot (-t^2)dt \\&= \frac{1}{2} \int (-t^2+t^4)dt \\&= \frac{1}{2} \left(-\frac{1}{3}t^3 + \frac{1}{5}t^5 \right) + C \\&= -\frac{t^3}{6} + \frac{t^5}{10} + C \\&= -\frac{(1-2x)^{\frac{3}{2}}}{6} + \frac{(1-2x)^{\frac{5}{2}}}{10} + C.\end{aligned}$$

$$(6) \int \frac{1}{\sqrt{e^x+1}} dx.$$

解: 设 $\sqrt{e^x+1}=t$, 则 $e^x+1=t^2$,

$$x = \ln(t^2-1), \quad dx = \frac{2t}{t^2-1} dt.$$

$$\begin{aligned}\text{于是 } \int \frac{1}{\sqrt{e^x+1}} dx &= \int \frac{1}{t} \cdot \frac{2t}{t^2-1} dt \\&= 2 \int \frac{1}{t^2-1} dt = 2 \cdot \left(\frac{1}{2} \right) \ln \left| \frac{t-1}{t+1} \right| + C\end{aligned}$$

$$= \ln \left| \frac{t-1}{t+1} \right| + C = \ln \frac{\sqrt{e^x+1}-1}{\sqrt{e^x+1}+1} + C.$$

§ 4—5

1. 利用积分表计算下列不定积分:

$$1. \int \frac{1}{x(2x+5)^2} dx.$$

解: 被积函数含有 $ax+b$,

在积分表(一)中查得公式(9),

$$\begin{aligned} & \int \frac{1}{x(ax+b)^2} dx \\ &= \frac{1}{b(ax+b)} - \frac{1}{b^2} \ln \left| \frac{ax+b}{x} \right| + C. \end{aligned}$$

现 $a=2$, $b=5$, 于是

$$\begin{aligned} & \int \frac{1}{x(2x+5)^2} dx \\ &= \frac{1}{5(2x+5)} - \frac{1}{25} \ln \left| \frac{2x+5}{x} \right| + C. \end{aligned}$$

$$2. \int \frac{dx}{5+3\cos x}.$$

解: 被积函数含有

$$a+b\cos x \quad (a^2 > b^2),$$

在积分表(十一)中查得公式(105),

$$\begin{aligned} & \int \frac{dx}{a+b\cos x} = \\ & \frac{2}{a+b} \sqrt{\frac{a+b}{a-b}} \arctan \left(\sqrt{\frac{a-b}{a+b}} \tan \frac{x}{2} \right) + C. \end{aligned}$$

现 $a=5$, $b=3$, 于是

$$\begin{aligned} & \int \frac{dx}{5+3\cos x} = \\ & \frac{2}{5+3} \sqrt{\frac{5+3}{5-3}} \arctan \left(\sqrt{\frac{5-3}{5+3}} \tan \frac{x}{2} \right) + C \end{aligned}$$

$$= \frac{1}{4} 2 \arctan \left(\frac{1}{2} \tan \frac{x}{2} \right) + C$$

$$= \frac{1}{2} \arctan \left(\frac{1}{2} \tan \frac{x}{2} \right) + C .$$

$$3. \int \frac{1}{\sqrt{(4-x^2)^3}} dx .$$

解：被积函数含有

$$\sqrt{a^2 - x^2} \quad (a > 0),$$

在积分表(八)中查得公式(60),

$$\int \frac{1}{\sqrt{(a^2 - x^2)^3}} dx = \frac{x}{a^2 \sqrt{a^2 - x^2}} + C .$$

现 $a = 2 > 0$, 于是

$$\int \frac{1}{\sqrt{(4-x^2)^3}} dx = \frac{x}{4\sqrt{4-x^2}} + C .$$

$$4. \int x^3 \sqrt{a^2 + x^2} dx \quad (a > 0, \text{为常数}).$$

解：这个积分不能在表中直接查到，需要进行变量代换.

$$\text{因为 } \int x^3 \sqrt{a^2 + x^2} dx = \frac{1}{2} \int x^2 \sqrt{a^2 + x^2} d(x^2),$$

令 $x^2 = u$, 于是

$$\begin{aligned} & \int x^3 \sqrt{a^2 + x^2} dx \\ &= \frac{1}{2} \int u \sqrt{a^2 + u} du = \frac{1}{2} \int u \sqrt{u + a^2} du . \end{aligned}$$

被积函数含有 $\sqrt{ku + b}$,

在积分表(二)中查得公式(11),

$$\begin{aligned} & \int u \sqrt{ku + b} du \\ &= \frac{2}{15k^2} (3ku - 2b) \sqrt{(ku + b)^3} + C . \end{aligned}$$

现 $k = 1$, $b = a^2$, $u = x^2$, 于是

$$\begin{aligned}& \int x^3 \sqrt{a^2 + x^2} dx \\&= \frac{1}{2} \left[\frac{2}{15 \cdot 1} (3x^2 - 2a^2) \sqrt{(a^2 + x^2)^3} + C \right] \\&= \frac{1}{15} (3x^2 - 2a^2) \sqrt{(a^2 + x^2)^3} + \frac{1}{2} C \\&= \frac{1}{15} (3x^2 - 2a^2) \sqrt{(a^2 - x^2)^3} + C_1. \\&\left(C_1 = \frac{1}{2} C \right)\end{aligned}$$